

Data Converters

Exercise 08 – Delta-Sigma ADC

Task I – Analysis of a Second-order Delta-Sigma ADC

The block diagram of a second-order discrete-time Delta-Sigma ADC is shown in Figure 1.

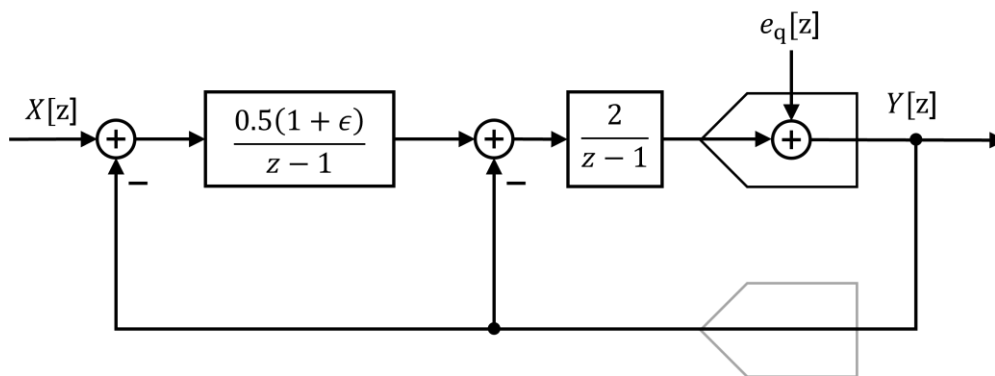


Figure 1: Second-order discrete-time Delta-Sigma ADC.

- Show that $\frac{y_{out}[z]}{x_{in}[z]} = 1/(z - 1)$ represents a discrete-time integrator.
- Determine the noise transfer function of the modulator in the z-domain.
- Apply the inverse z-transform, i.e., $z = e^{s/f_s} = e^{i2\pi f/f_s}$, to transfer your result to the s-domain.
- Simplify your result by assuming $f/f_s \ll 1$ and $e^x \approx 1 + x$ for $x \ll 1$.
- Which error is modeled by the term $(1 + \epsilon)$?
- How much does the in-band quantization noise increase for $\epsilon = -0.5$?

Task II – Analysis of a Multi-bit ADC / Single-Bit DAC Delta-Sigma ADC

The Delta-Sigma ADC shown in Figure 2 uses a B-bit ADC for increased resolution and a 1-bit DAC to avoid distortion due to DAC nonlinearity. The block *MSB* outputs the most-significant bit of the B-bit ADC.

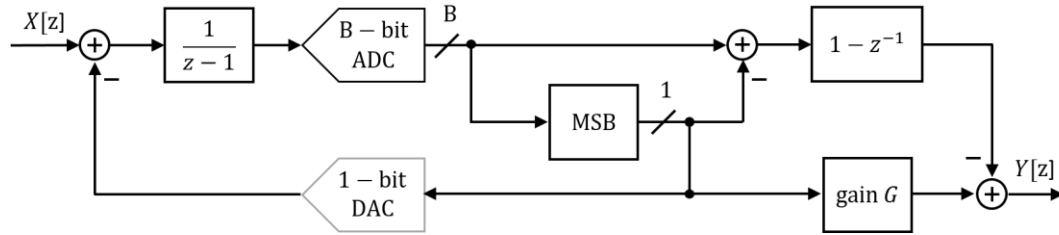


Figure 2: Multi-bit ADC / single-bit DAC Delta-Sigma ADC.

- Draw the linearized model of the Delta-Sigma ADC assuming that the quantization error $e_q[z]$ of the B-bit quantizer and the truncation error $e_t[z]$ of the block *MSB* can be modelled as additive white noise sources.
- Determine the signal transfer function, the quantization error transfer function, and the truncation error transfer function.
- Considering your result for the truncation error transfer function, what is the optimal gain G ?
- Assuming that the gain G is set to its optimal value, state the *SQNR* of this modulator as a function of the resolution B and the oversampling ratio *OSR*.
- How much larger is the truncation error compared with the quantization error?
- Comment on the potential limitation of this architecture.

Note: Determine the truncation error transfer function as a function of the noise transfer function (implemented in the analog domain) and the filter function $1 - z^{-1}$ (implemented in the digital domain).